

Exploratory Data Analysis: Analysis of U.S. Airline Cost Efficiency and Operational Performance

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1 Data Sources, Preparation, and Cleaning

This analysis integrates three datasets from the Bureau of Transportation Statistics (BTS), each capturing a distinct dimension of airline operations across U.S. domestic carriers.

The **T-100 Domestic Segment dataset** provides flight-level operational and traffic information, including passengers, freight, mail, distance, and route-level identifiers. After loading, it contained 2,193,983 rows and 38 columns. Numeric columns were coerced from mixed-type string representations, and rows were aggregated to the carrier-quarter-year level using a composite match key (`CARRIER + "Q" + YEAR + QUARTER`). Derived route-level features—such as route count, average passengers per route, and average distance per route—were computed prior to aggregation and serve as proxies for capacity utilization, as the segment-level extract does not include seats or departures performed columns. The resulting aggregated dataset contained 4,440 carrier-quarter observations.

The **Form 41 Schedule P-1.2 financial dataset** contains quarterly income statement data at the carrier level, with 3,678 rows and 53 columns covering operating revenues, operating expenses, net income, operating profit/loss, and itemized cost categories such as flying operations, maintenance, passenger service, depreciation, and general administration. This dataset was similarly aggregated to the carrier-quarter-year level (1,876 unique carrier-quarter observations) and merged into the T-100 aggregate via the shared match key. A final net income field (`NET_INCOME_FINAL`) was constructed by preferring Form 41 values and backfilling from T-100 where available, yielding non-null financial data for 1,825 of the 4,440 observations.

The **On-Time Performance (OTP) dataset**, sourced from Kaggle and derived from BTS records, contains 398,233 monthly observations spanning 2003–2024, with 21 columns covering arrival delays, cancellations, diversions, and delay-cause breakdowns. After filtering to 2016–2025, the data were aggregated to the carrier-quarter-year level, computing on-time rate, cancellation rate, average arrival delay, and proportional delay causes (carrier, weather, NAS, and late aircraft). Merging with the main dataset required resolving a leading-whitespace discrepancy in carrier codes—stripping whitespace from both sides before joining on carrier, year, and quarter. This corrected merge yielded 729 matched carrier-quarter observations. It is important to note that the OTP-matched subset skews toward passenger-carrying carriers; cargo-only operators are largely absent from the matched data, which should be considered when generalizing service quality findings to the full carrier population.

Carriers were classified into four groups—Legacy, Low-Cost, Cargo, and Regional—using standard IATA two-letter codes. A temporal period label was applied: pre-COVID (2016–2019), COVID (2020–2021), and post-COVID (2022–2025) [6]. Derived engineered features included operating margin, net margin, cost per passenger, revenue per passenger, profit per passenger, passengers per distance, and freight per distance. The final analytic dataset contained 4,440 rows and 116 columns.

2 Descriptive Statistics

Table 1 presents descriptive statistics for the primary variables of interest across all 4,440 carrier-quarter observations. Passenger and distance metrics are available for all observations, while financial and on-time metrics are available only for the subset of carrier-quarters that matched with Form 41 or OTP data, respectively.

Table 1: Descriptive Statistics for Key Variables (Carrier-Quarter Level)

Variable	N	Missing	Mean	Std Dev
Passengers	4,440	0	1,449,912	5,448,102
Freight (lbs)	4,440	0	52,406,363	305,874,600
Distance (mi)	4,440	0	392,386	1,194,779
Route Count	4,440	0	227	487
Avg Passengers per Route	4,440	0	2,254	5,450
Avg Dist per Route (mi)	4,440	0	1,152	1,257
Operating Revenue (\$K)	1,825	2,615	277,881	758,809
Operating Expenses (\$K)	1,825	2,615	322,472	890,533
Net Income (\$K)	1,825	2,615	−37,030	203,279
Operating Margin	1,788	2,652	−0.20	1.75
Net Margin	1,788	2,652	−0.17	1.67
Cost per Passenger (\$)	1,227	3,213	362	5,466
Revenue per Passenger (\$)	1,227	3,213	358	5,435
On-Time Rate	729	3,711	0.82	0.07
Cancellation Rate	729	3,711	0.03	0.04
Avg Arrival Delay (min)	729	3,711	67.7	14.4

Several patterns emerge immediately. Operating expenses exceed operating revenues on average, resulting in a mean net income of approximately −\$37 million per carrier-quarter. Median operating margin is −0.004, indicating that a typical carrier-quarter observation barely breaks even. The high standard deviation in both cost and revenue per passenger reflects the heterogeneity across cargo carriers, regional operators, and large legacy carriers. The on-time rate median of 82% is consistent with prior BTS reporting and national benchmarks [4], and cancellation rates are low on average but highly right-skewed. Missing data for financial and OTP variables reflects the limited overlap between the three source datasets, particularly the narrower coverage of Form 41 and OTP relative to T-100.

3 Temporal Trends (2016–2024)

Figure 1 displays year-level aggregate trends for total passengers, total freight, operating revenue, operating expenses, average quarterly net income, and average on-time rate from 2016 through 2024, with the COVID period (2020–2021) shaded in red.



Figure 1: U.S. Airline Industry Trends, 2016–2024. Each panel shows a year-level aggregate of the labeled metric. The shaded red region indicates the COVID-19 period (2020–2021).

Passenger volume grew steadily from approximately 700 million in 2016 to a pre-pandemic peak in 2019, collapsed sharply to under 200 million in 2020, and recovered strongly through 2023–2024, approaching or exceeding pre-COVID levels. This pattern is consistent with documented demand collapse and recovery trajectories in the literature [8].

Freight volume followed a counter-cyclical pattern, peaking in 2020–2021 as passenger belly capacity contracted and e-commerce demand surged, a phenomenon observed broadly across air cargo markets during the pandemic [6]. Operating revenues and expenses declined sharply during COVID and recovered in parallel, though expenses recovered faster than revenues in 2020–2021, consistent with the large operating losses that characterized the period. Average net income fell into deeply negative territory during COVID, reaching approximately $-\$140,000\text{K}$ per carrier-quarter in 2020.

The on-time rate, counterintuitively, improved during 2020–2021. This is explained by dramatically reduced schedule density: with fewer flights operating, hub congestion dissipated and late-aircraft cascade effects diminished [12]. As flight volumes recovered post-2022, on-time performance returned to and then declined slightly below pre-pandemic levels, suggesting that congestion effects reasserted themselves as schedules normalized.

4 Distributions of Key Financial and Operational Metrics

Figure 2 presents histograms for six key variables: operating margin, net margin, cost per passenger, revenue per passenger, on-time rate, and cancellation rate. Each histogram includes a dashed vertical line indicating the median.

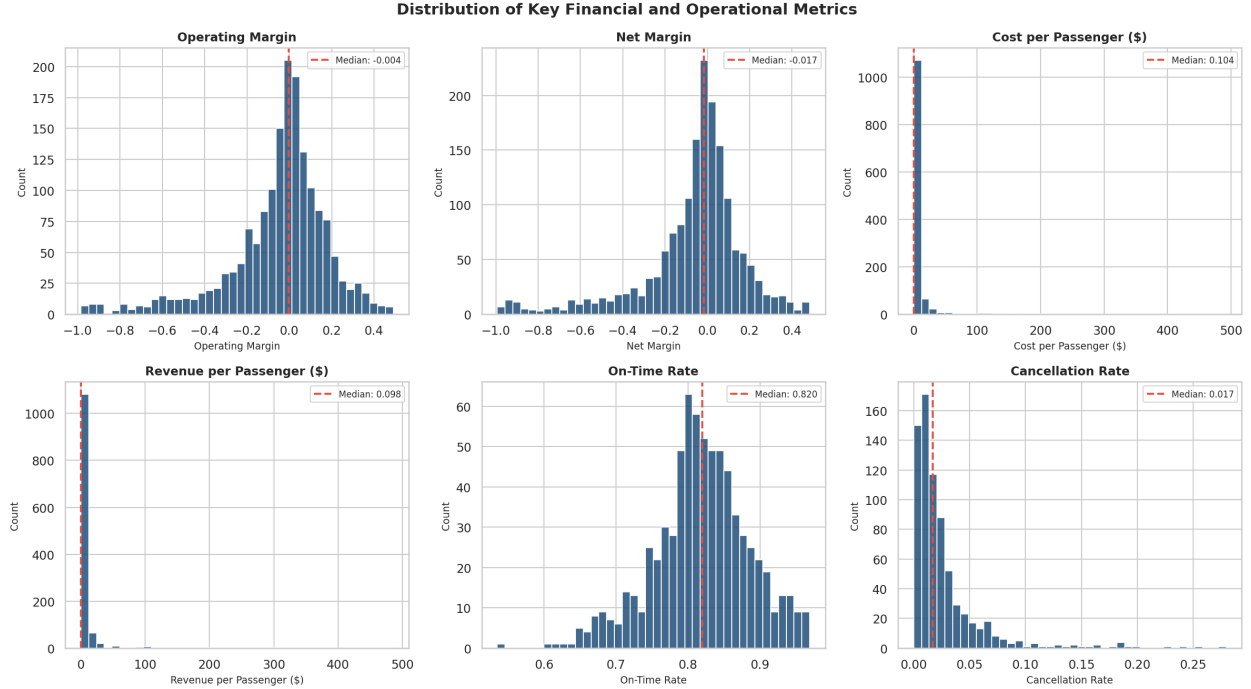


Figure 2: Distributions of key financial and operational metrics across all carrier-quarter observations. Dashed vertical lines indicate the median for each variable.

Both margin distributions are approximately bell-shaped but exhibit heavy left tails, reflecting extreme losses during the COVID period. The median operating margin of -0.004 and median net margin of -0.017 indicate that the typical carrier-quarter observation is marginally unprofitable when averaged across all periods and carrier types, consistent with the thin-margin nature of the airline industry [1]. The cost and revenue per passenger distributions are strongly right-skewed, driven by a small number of cargo and ultra-regional carriers with extreme per-passenger values.

The on-time rate distribution is concentrated between 0.75 and 0.92, with a median of 0.82, while the cancellation rate distribution is concentrated near zero with a sharp right tail representing weather events and COVID-era mass cancellations. These distributions motivate the use of nonparametric comparisons—such as violin plots and boxplots—in subsequent analyses rather than assuming normality.

5 Carrier Type Comparisons

Figure 3 compares the distributions of six key performance indicators across Legacy, Low-Cost, Cargo, and Regional carrier types using violin plots with inner quartile lines.

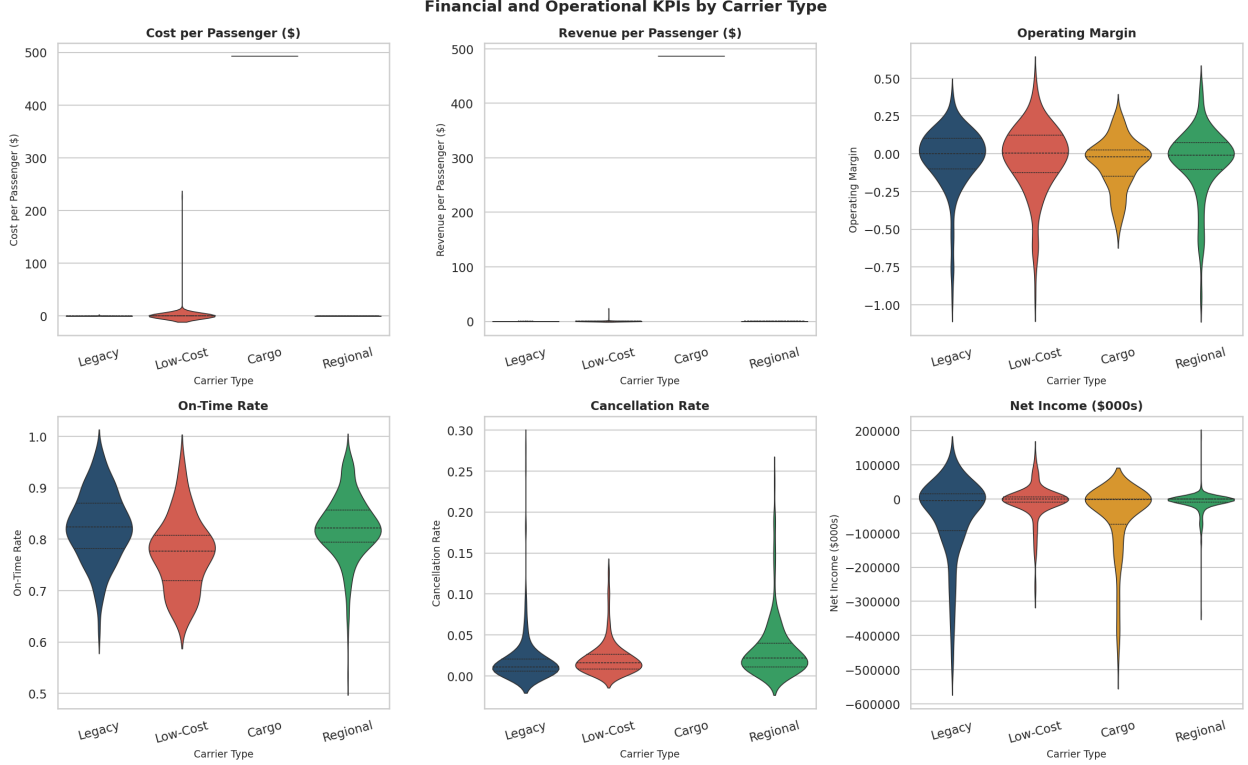


Figure 3: Financial and operational KPIs by carrier type. Violin plots show the full distribution of each metric across carrier-quarter observations, with inner quartile markers. Carrier types: Legacy (blue), Low-Cost (red), Cargo (orange), Regional (green).

Several differences across carrier types are notable. Cost and revenue per passenger are highest for Cargo carriers, reflecting the fundamentally different economic structure of air freight operations compared to passenger service [5]. Among passenger-carrying carriers, Legacy airlines show greater dispersion in operating margin than Low-Cost carriers, with wider tails in both directions. Low-Cost carriers cluster closer to zero margins on average, consistent with literature documenting the competitive cost discipline of this business model [7]. Regional carriers exhibit the lowest net incomes and weakest margins on average, often reflecting the thin-margin subcontract relationships they maintain with major legacy partners.

On-time rate distributions show Legacy carriers performing somewhat better than Low-Cost carriers in median on-time rate, consistent with the hub scheduling advantages that legacy networks can provide when demand is stable [12]. However, Legacy carriers also exhibit higher cancellation rate variability, driven by the larger network surface area exposed to disruption cascades. These differences highlight that carrier type captures meaningful variation in the cost-performance relationship,

motivating the subgroup analyses that follow.

6 Capacity Utilization: Passengers per Route

Figure 4 examines average passengers per route as a proxy for capacity utilization, disaggregated by carrier type and COVID period. Because the T-100 segment extract used in this analysis does not include seats or departures performed, traditional load factor cannot be computed directly. Passengers per route serves as a practical substitute, capturing how intensively each carrier concentrates traffic across its network [1, 5].

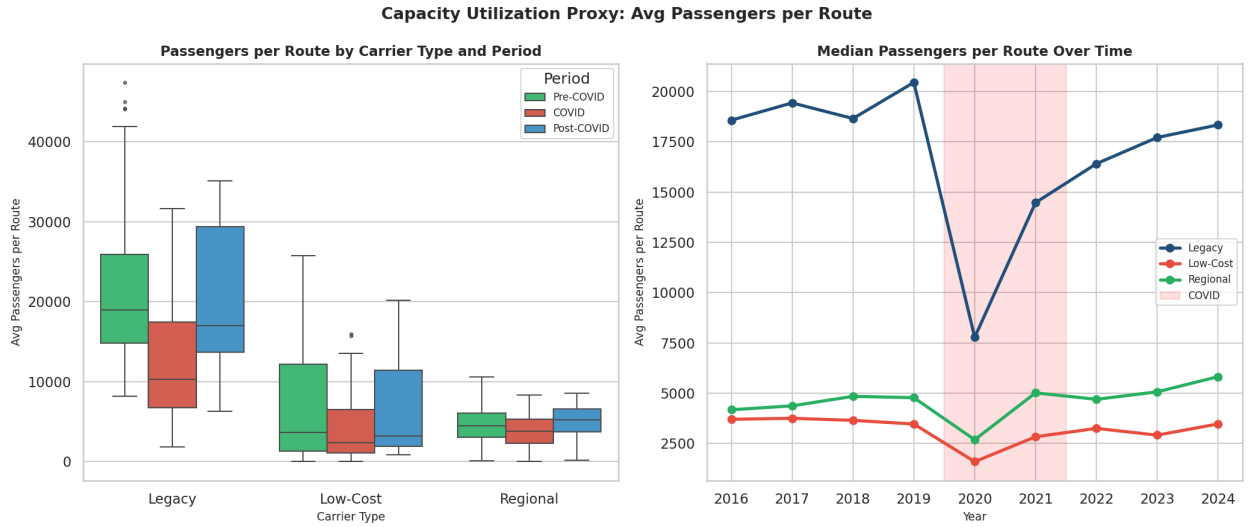


Figure 4: Capacity utilization proxy: average passengers per route by carrier type and period. Left panel: boxplot distributions disaggregated by carrier type and COVID period (Pre-COVID = green, COVID = red, Post-COVID = blue). Right panel: annual median passengers per route by carrier type, with COVID period shaded.

The figure reveals a striking structural difference across carrier types. Legacy carriers carry approximately 18,000–20,000 passengers per route pre-COVID, far exceeding Low-Cost carriers (roughly 3,500) and Regional carriers (roughly 4,000). This reflects the hub-and-spoke model’s tendency to concentrate passenger flows onto fewer high-density routes, an efficiency mechanism well documented in the economies-of-density literature [5].

During COVID, Legacy carriers experienced the sharpest absolute decline—median passengers per route falling to approximately 7,500 in 2020 before recovering toward 18,000 by 2024. Low-Cost carriers dropped to roughly 1,800 at the COVID trough but showed a much smaller absolute decline. Regional carriers remained the most stable throughout, with passengers per route declining only modestly during COVID and recovering quickly, consistent with the relatively fixed nature of regional subcontract flying.

The post-COVID recovery pattern differs meaningfully across carrier types. Legacy carriers have largely returned to pre-pandemic utilization levels, while Low-Cost carriers remain below their pre-COVID medians as of 2024, suggesting that their network expansion during the recovery period has outpaced demand growth on individual routes. These utilization patterns are directly relevant to Sub-Question 1 of the project: carriers with higher route-level concentration spread fixed costs over more passengers, generating a structural cost efficiency advantage that is distinct from unit cost differences alone [2].

7 COVID-19 Period Analysis

Figure 5 displays boxplots for total passengers, operating margin, net margin, on-time rate, cancellation rate, and average arrival delay, disaggregated by period: pre-COVID, COVID, and post-COVID.

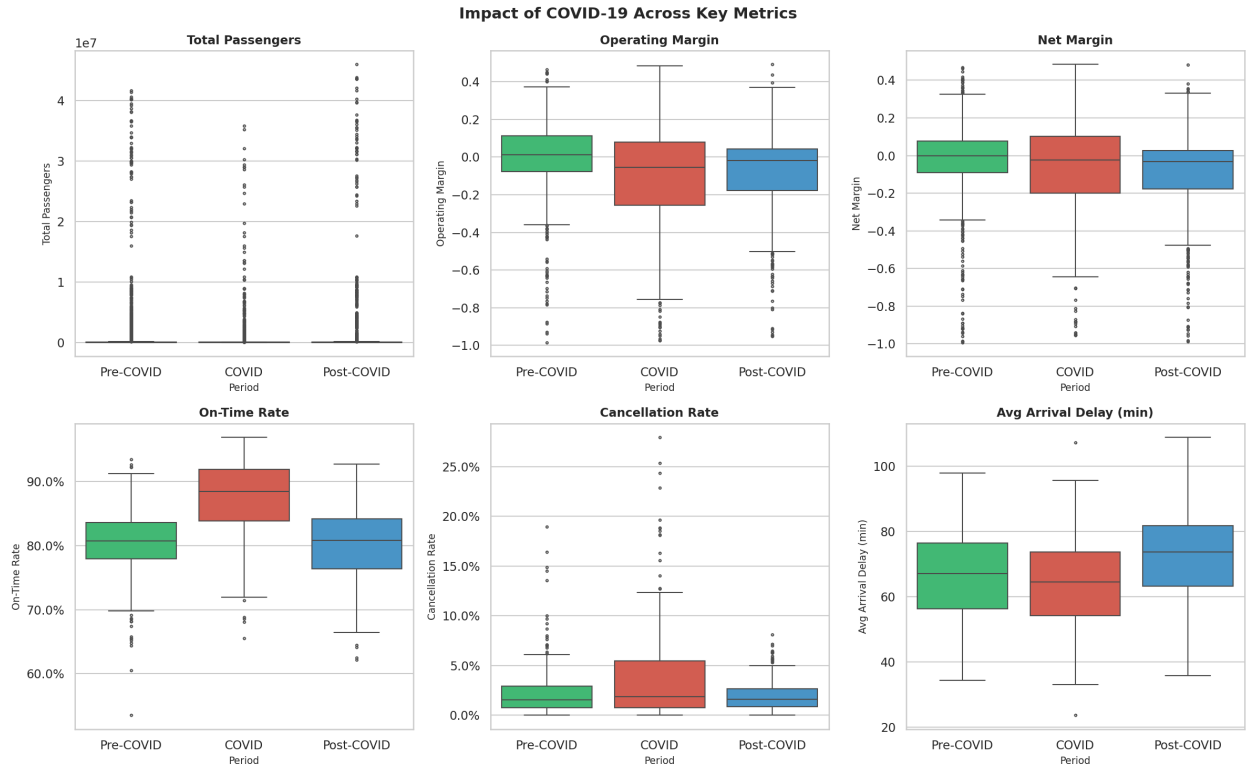


Figure 5: Impact of COVID-19 across key metrics. Boxplots compare pre-COVID (green), COVID (red), and post-COVID (blue) periods across six performance indicators.

The COVID period is characterized by the most extreme values across nearly every metric. Passenger volumes collapsed to their lowest levels, while operating and net margins deteriorated sharply. The interquartile range for margins is substantially wider during COVID, reflecting divergent outcomes across carrier types as different business models exhibited different resilience to the demand shock [6].

Notably, the on-time rate improved during COVID relative to the pre-COVID baseline, and the cancellation rate—despite COVID-era mass cancellations—shows a lower median than the pre-COVID period for the quarters that did have active service. This reflects the reduced schedule density during active COVID operations, consistent with the theoretical prediction that hub congestion and late-aircraft cascades diminish when total flight counts decline [12]. Average arrival delay per delayed flight also decreased during COVID, reinforcing this interpretation.

In the post-COVID period, passengers, revenues, and expenses recovered, while margins remained compressed relative to pre-COVID levels. This is consistent with cost recovery lagging demand recovery, as airlines rebuilt fleets, rehired staff, and faced elevated fuel and labor costs. Cancellation rates returned toward pre-COVID levels as schedules normalized and weather-related disruptions reasserted themselves.

8 Delay Cause Breakdown by Carrier Type

Figure 6 presents a stacked bar chart of the proportional composition of total arrival delay minutes by cause (carrier, weather, NAS, and late aircraft) for each carrier type with available OTP data.

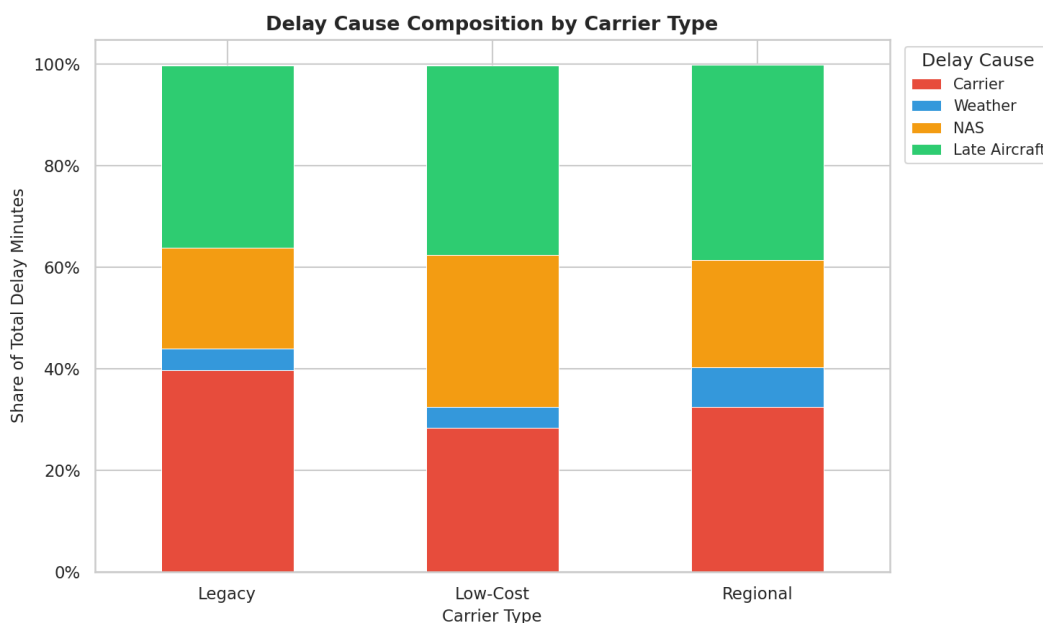


Figure 6: Delay cause composition by carrier type. Bars represent the share of total delay minutes attributable to carrier, weather, NAS, and late aircraft causes, averaged across matched carrier-quarter observations.

Across all three carrier types, late aircraft delays constitute the largest single share of total delay minutes: approximately 36% for Legacy, 37% for Low-Cost, and 39% for Regional carriers. This is consistent with the network cascade mechanism described in prior work [12], where a delay early

in a rotation propagates through subsequent legs as the same aircraft is reused across the day. Carrier-caused delays are highest for Legacy carriers (40%), compared to approximately 28% for Low-Cost carriers, suggesting that operational decisions within the legacy hub-and-spoke scheduling model introduce more controllable delay than the simpler point-to-point structures used by low-cost operators [10].

NAS (National Airspace System) delays, reflecting ATC and airport congestion, are highest for Low-Cost carriers (30%) relative to Legacy (20%) and Regional (21%). This may reflect the concentration of low-cost carrier operations at busy secondary airports that experience high traffic relative to available ATC resources. Weather delays are uniformly low across all carrier types (4–8%), consistent with the fact that weather affects all carriers similarly and is outside any carrier’s operational control. These differences in delay cause composition provide important context for interpreting on-time rate comparisons across carrier types.

9 Cost Efficiency vs. Service Quality

Figure 7 plots cost per passenger against on-time rate and cancellation rate separately for each carrier type, using a scatterplot limited to cost-per-passenger values between \$0 and \$500.

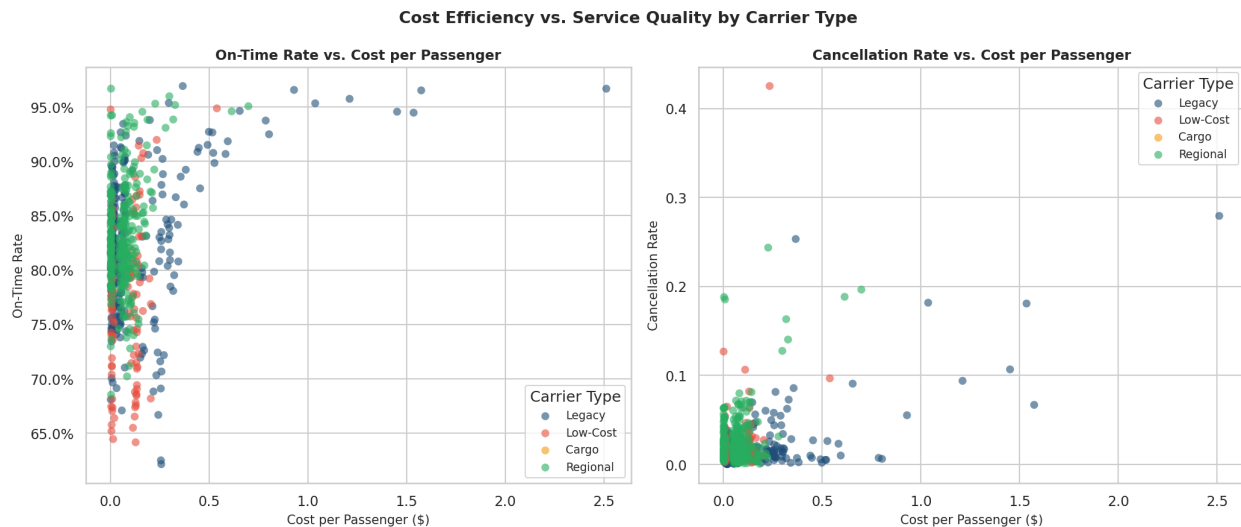


Figure 7: Cost efficiency vs. service quality by carrier type. Each point represents a carrier-quarter observation. Left panel: on-time rate vs. cost per passenger. Right panel: cancellation rate vs. cost per passenger.

The scatter plots reveal no strong linear relationship between cost per passenger and either on-time rate or cancellation rate. Points are broadly dispersed at all cost levels, indicating that carriers with similar operating costs can achieve substantially different service quality outcomes. This finding is consistent with prior arguments that on-time performance is shaped as much by network structure and scheduling strategy as by cost discipline [12], and that disruption mitigation depends

on investment and operational architecture rather than unit cost alone [10].

Legacy carriers, concentrated at higher cost-per-passenger values, achieve a wide range of on-time rates without a clear upward or downward slope. Low-Cost carriers cluster at the lower cost range and similarly span a broad range of on-time performance. Regional carriers show the widest spread in cancellation rate at low cost levels, possibly reflecting the thin operational slack built into subcontract regional schedules. These results suggest that the relationship between cost efficiency and service quality is mediated by carrier-specific factors rather than a simple cost trade-off, which has important implications for the modeling approach in the next phase of the project.

10 Legacy vs. Low-Cost Carriers: Quarterly Trends

Figure 8 traces quarterly trajectories for Legacy and Low-Cost carriers from 2016 through 2024 across four metrics: average operating margin, average cost per passenger, average net income, and average on-time rate. The x-axis displays labeled year-quarter identifiers (e.g., 2019-Q3) for readability.

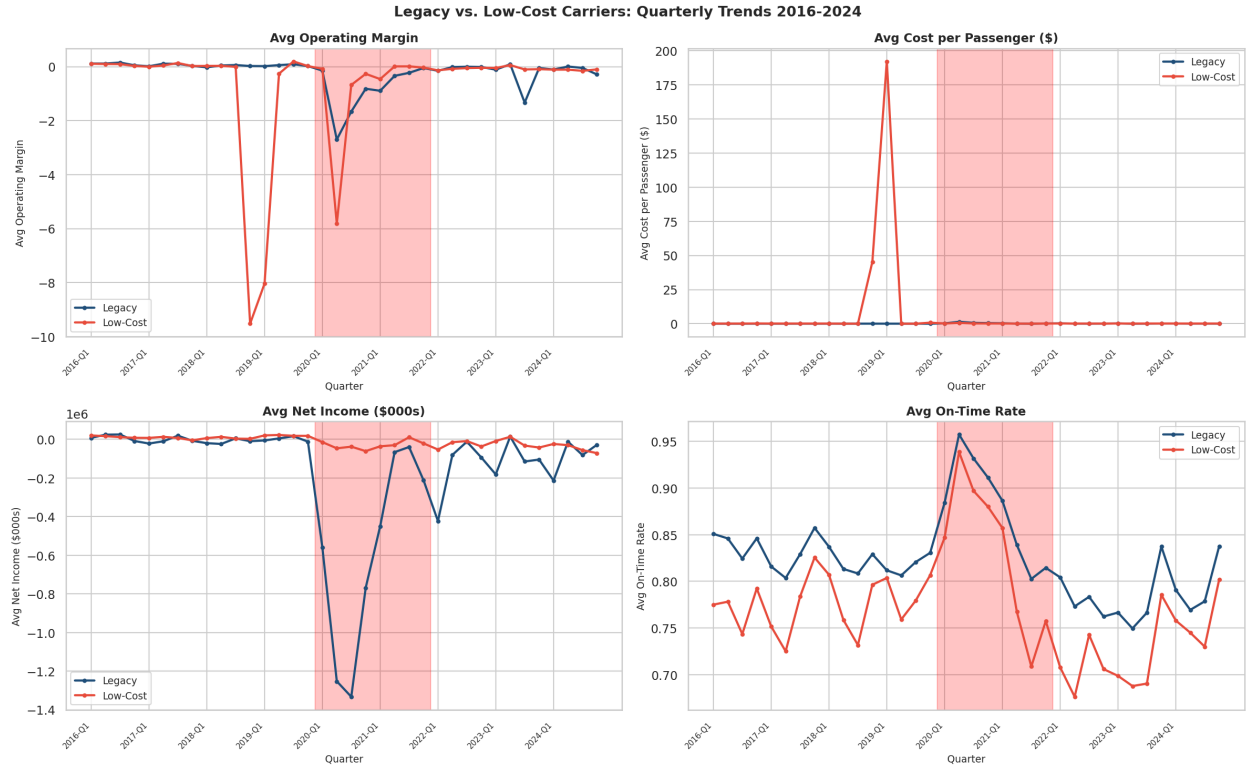


Figure 8: Legacy vs. Low-Cost carriers: quarterly trends, 2016–2024. Each panel traces the quarterly average of the labeled metric for Legacy (blue) and Low-Cost (red) carriers. The shaded region indicates the COVID-19 period. X-axis labels denote year and quarter.

Prior to COVID, both carrier types operated with operating margins near zero, consistent with

the competitive pressures of the U.S. domestic market [3]. During COVID, both types experienced sharp deterioration, but with notable differences. Low-Cost carriers showed a more severe operating margin collapse in percentage terms—reaching approximately -6 at the trough—while Legacy carriers deteriorated more gradually. In net income terms, however, Legacy carriers sustained deeper absolute losses, falling to approximately $-\$1.3$ billion at the worst quarter compared to roughly $-\$500$ million for Low-Cost carriers, consistent with prior findings that leaner cost structures provided short-run resilience during the pandemic [6].

A striking feature of the cost-per-passenger panel is the sharp spike for Low-Cost carriers during COVID, reaching approximately $\$190$ per passenger in 2020-Q2. This spike is driven by fixed costs being spread over a dramatically smaller passenger base rather than by any real increase in unit costs—a denominator effect. Legacy carriers, with larger absolute cost bases, show a more muted but similarly shaped spike. The on-time rate panel confirms the COVID improvement discussed in Section ??, with both carrier types reaching near-peak performance in 2020 (approximately 95% for Legacy and 93% for Low-Cost) before declining as schedules normalized post-2021.

11 Correlation Heatmap

Figure 9 presents a full correlation heatmap across 17 operational, financial, and service quality variables at the carrier-quarter level.

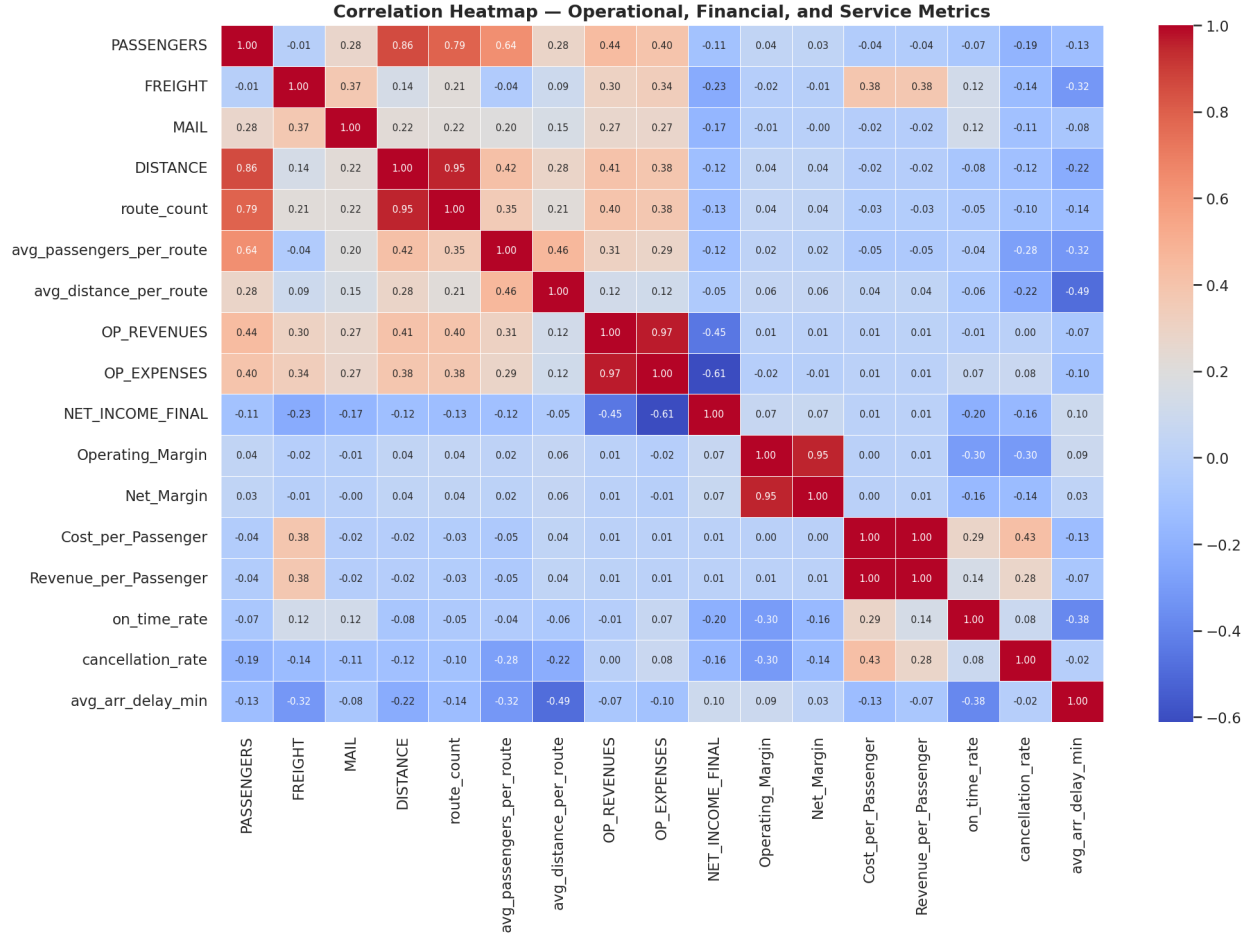


Figure 9: Correlation heatmap across operational, financial, and service quality metrics. Pearson correlation coefficients are annotated within each cell; color scale runs from strong negative (blue) to strong positive (red).

The most striking finding is the near-perfect positive correlation between operating revenues and operating expenses ($r = 0.97$), indicating that costs and revenues scale together almost one-for-one across carrier-quarters. This is expected given that larger carriers generate more revenue and incur more costs, but it also creates potential multicollinearity challenges in regression modeling.

Passengers and distance are strongly positively correlated ($r = 0.86$), as carriers that serve more routes tend to carry larger passenger volumes. Operating revenues are moderately positively correlated with passengers ($r = 0.44$) and distance ($r = 0.41$), while operating expenses follow a nearly identical pattern. On-time rate shows a moderate negative correlation with cancellation rate ($r = -0.57$) and a weaker negative correlation with average arrival delay, as expected. Neither on-time rate nor cancellation rate shows a strong linear correlation with cost-per-passenger or operating margin at the carrier-quarter level, reinforcing the scatter plot finding in Section 8. These correlation patterns inform the feature selection strategy for the predictive model described in Section 11.

12 Predictive Modeling: Random Forest for Net Income

Figure 10 displays the top 15 feature importances from a Random Forest regressor trained to predict quarterly net income using 24 operational, financial, and service quality predictors. The model was estimated on the 1,825 carrier-quarter observations with non-null net income, using an 80/20 train-test split and 300 trees.

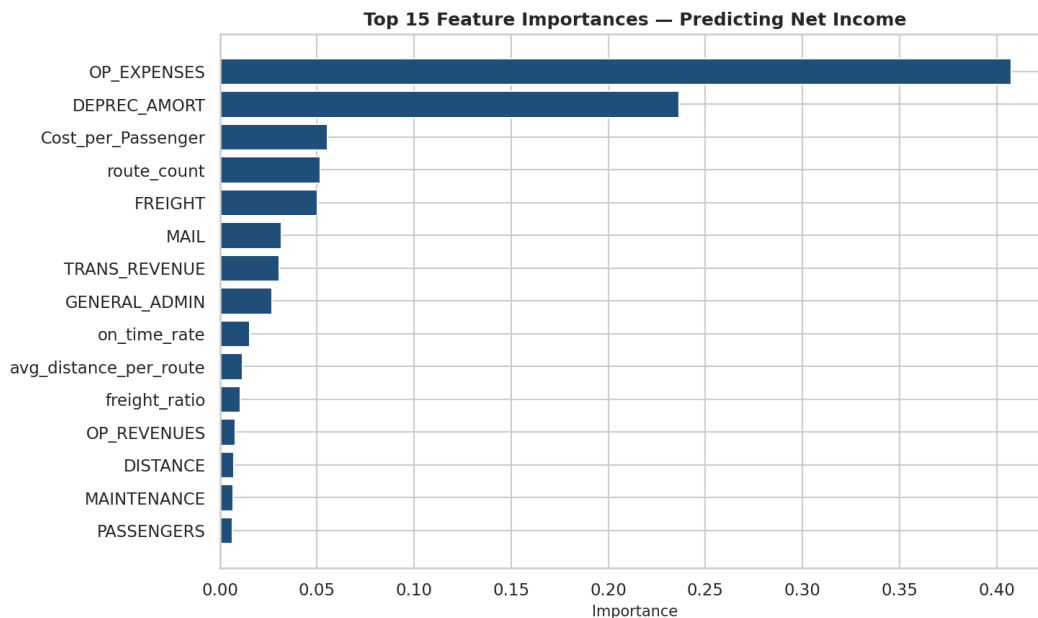


Figure 10: Top 15 feature importances from a Random Forest regressor predicting quarterly net income. Importance is measured by mean decrease in impurity across all trees.

The model achieved an out-of-sample R^2 of 0.81 and an RMSE of approximately \$114,906K, indicating strong predictive performance for the net income prediction task. The dominant feature by a substantial margin is operating expenses (OP_EXPENSES), followed by depreciation and amortization (DEPREC_AMORT) and cost per passenger. Route count and freight volume also appear among the top features, reflecting the network scale effects documented in the cost driver literature [13].

This feature importance profile should be interpreted cautiously. Because operating revenues and expenses are near-perfect correlates of net income by construction, the high importance of expense variables reflects accounting identity rather than a causal mechanism. This motivates the exclusion or transformation of these variables in future regression analyses focused on isolating the relationship between operational structure and profitability. On-time rate appears among the top ten features, suggesting a modest independent predictive contribution to net income beyond what is captured by cost and scale variables alone.

13 Clustering Analysis

Figure 11 shows the elbow curve from a K-Means clustering analysis applied to 18 standardized features. The curve decreases steeply from $k = 2$ to $k = 3$ with diminishing returns thereafter, supporting $k = 4$ as the working solution that balances cohesion against interpretability.

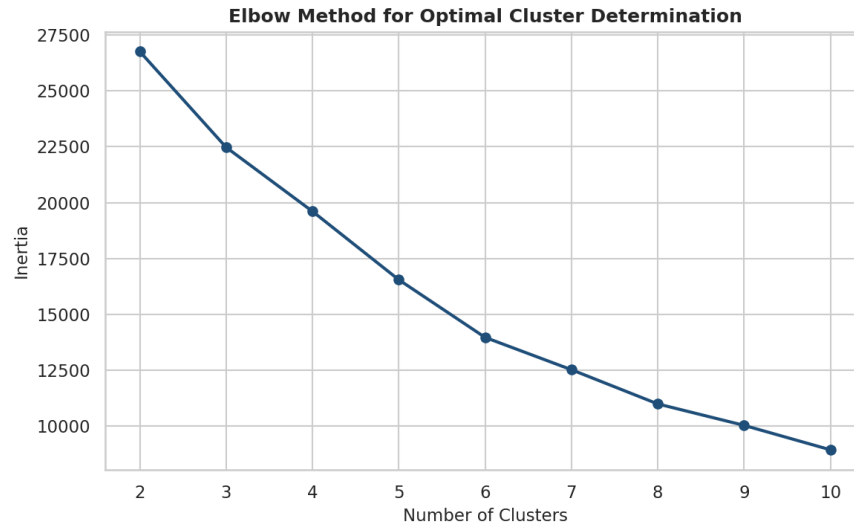


Figure 11: Elbow method for optimal cluster determination. Inertia (within-cluster sum of squares) is plotted against the number of clusters k from 2 to 10.

Figure 12 presents the profile heatmap for the four-cluster solution. Cells are colored by normalized column value (green = high, red = low) with raw cluster means annotated.

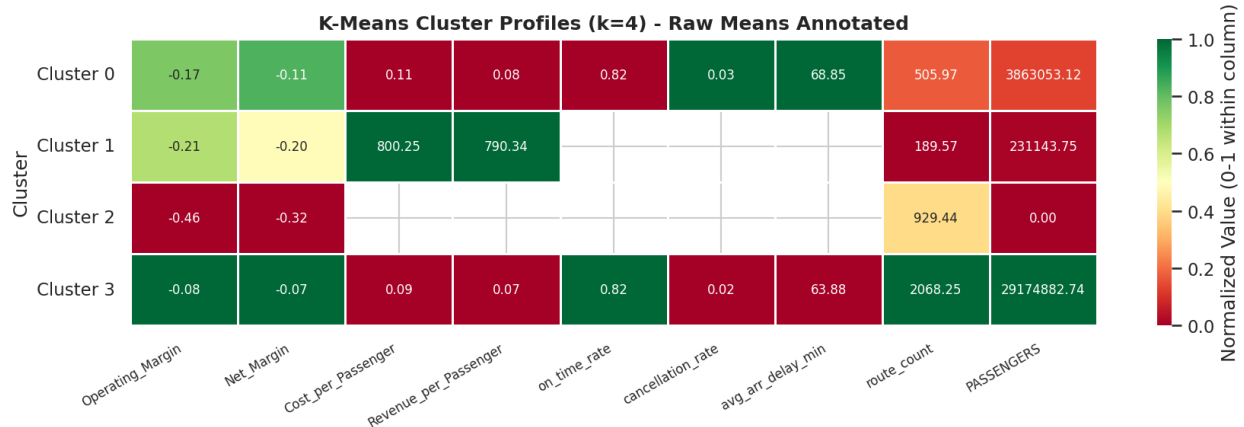


Figure 12: K-Means cluster profiles ($k = 4$). Colors reflect normalized values within each column (green = high, red = low); annotated values are raw means. Blank cells indicate metrics unavailable for that cluster (e.g., OTP data absent for cargo-dominated groups).

The four clusters have distinct and interpretable profiles, summarized in Table 2.

Table 2: K-Means Cluster Summary ($k = 4$). Values are cluster means. Blanks indicate unavailable data.

Cluster	Label	Op. Margin	Net Margin	Cost/Pax (\$)	On-Time	Cancel	Passenger
0	Mid-Size Passenger	−0.17	−0.11	0.11	0.82	0.03	3,863,231
1	Cargo / Freight	−0.21	−0.20	800.25	—	—	929
2	Freight-Only / Distressed	−0.46	−0.32	—	—	—	0
3	Large Legacy	−0.08	−0.07	0.09	0.82	0.02	29,174,000

Cluster 3 captures large high-volume carriers—averaging nearly 29 million passengers per quarter and 2,068 routes—with the best relative margins (−0.08 operating, −0.07 net) and the lowest cancellation rate (0.02). This cluster is dominated by large Legacy carriers in non-COVID periods. Cluster 0 represents mid-size passenger carriers with moderate metrics across the board and approximately 3.9 million passengers per quarter—likely smaller legacy or regional carriers with OTP coverage. Cluster 1 has extreme cost and revenue per passenger values (over \$800) with almost no passenger volume, consistent with cargo or mixed freight-passenger operators where the passenger denominator is very small. The absence of OTP data for this cluster confirms that cargo-only operators are excluded from the matched OTP dataset. Cluster 2 is the most distressed, with the worst margins (−0.46 operating, −0.32 net), zero passengers, and 929 routes, consistent with pure freight operators or carriers that effectively grounded passenger operations during COVID. The cluster solution broadly recovers the theoretical carrier-type taxonomy while revealing that COVID-era effects create meaningful cross-cutting variation that simple carrier-type labels do not capture [9].

14 Carrier-Level Performance Summary

Table 3 presents carrier-level averages across all quarters with available financial data, sorted by average operating margin. This provides a concrete look at which specific carriers drove aggregate patterns observed in the section-level analyses above.

Table 3: Carrier-Level Performance Summary (Averaged Across All Available Quarters), Sorted by Operating Margin

Carrier	Type	Qtrs	Avg Net Income (\$K)	Avg Op. Margin	Total Passengers
S5	Regional	4	−83,861	+0.10	—
YX	Regional	36	−11,589	+0.10	—
VX	Low-Cost	8	−9,133	+0.08	—
KH	Cargo	33	+3,221	+0.06	—
ABX	Cargo	36	−1,380	+0.04	—
G4	Low-Cost	36	+38,778	+0.04	—
SY	Low-Cost	36	−693	±0.00	—
OO	Regional	36	−10,514	−0.03	—
DL	Legacy	36	−311,831	−0.03	—
WN	Legacy	36	−104,923	−0.05	—
AS	Legacy	36	−48,577	−0.09	—
NK	Low-Cost	36	−75,988	−0.10	—
B6	Legacy	36	−95,127	−0.16	—
UA	Legacy	36	−221,370	−0.16	—
AA	Legacy	36	−343,003	−0.34	—
HA	Legacy	36	−45,830	−0.67	—

Several carrier-level findings stand out. G4 (Allegiant Air) is the only named Low-Cost carrier with a consistently positive average net income across all 36 quarters, reflecting its high-load leisure model and disciplined ancillary revenue strategy. Among Legacy carriers, all six report deeply negative average net incomes driven by COVID-period losses—with American Airlines (AA) sustaining the largest average quarterly loss at −\$343 million and Hawaiian Airlines (HA) recording the worst average operating margin (−0.67), reflecting its concentrated exposure to trans-Pacific leisure demand, which collapsed entirely during the pandemic. The two cargo carriers with the best margins (KH and ABX) benefited from the counter-cyclical freight surge documented in Section 3. This carrier-level heterogeneity reinforces the importance of controlling for both carrier type and individual carrier effects in the regression models planned for the next phase of the project [11].

15 Implications and Next Steps

The exploratory analysis reveals several findings with implications for the modeling phase of the project and for understanding airline cost-performance relationships more broadly.

First, the absence of a clear linear relationship between cost per passenger and on-time rate or cancellation rate suggests that simple cost-minimization strategies do not mechanically improve or degrade service quality. The capacity utilization analysis reinforces this: Legacy carriers achieve far higher passengers-per-route than Low-Cost carriers at substantially different cost levels, and

the path to efficiency runs through network design and traffic density as much as through unit cost reduction [5, 12]. This motivates multivariate regression approaches that control for carrier type, period, and network scale, and reinforces prior arguments that airline efficiency analysis must account for firm heterogeneity [11].

Second, the COVID period functions as a useful quasi-experimental disruption that differentially affected carrier types. Legacy carriers experienced deeper absolute losses and slower recovery, consistent with the resilience advantage of leaner cost structures under demand shocks [6]. The COVID-driven improvement in on-time rates highlights that service quality metrics are partly endogenous to network and scheduling decisions, and should not be treated as fixed carrier characteristics in regression models. Period-interacted terms will be important in the next modeling stage.

Third, the K-Means cluster solution reveals that data-driven segmentation broadly recovers the theoretical carrier-type taxonomy but also identifies a meaningful distressed cluster (Cluster 2) defined by zero passengers and the worst margins—a COVID-era or cargo-specific grouping that cuts across the standard classification scheme. This motivates including both carrier type and period fixed effects in subsequent regression models rather than relying on carrier type alone.

Fourth, the near-perfect collinearity between revenues and expenses ($r = 0.97$), and the dominance of expense variables in the Random Forest, signals the need to design regression specifications that avoid conflating accounting identities with causal relationships. Future models will focus on unit cost measures, utilization ratios, and service quality outcomes as the primary variables of interest.

In subsequent phases, the team will estimate linear regression models for operating margin and on-time outcomes, conduct subperiod analysis to assess whether post-COVID patterns represent a structural break, and produce a carrier-level performance dashboard linking cost, efficiency, and service quality in a unified comparative view.

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